

**ARTIFICIAL NEURAL NETWORK FRAMEWORK FOR ENSO-
DRIVEN GLOBAL
LAND TEMPERATURE ANOMALY FORECASTING AND
INTERACTIVE
3D CLIMATE VISUALIZATION (1950–2035)**

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CERTIFICATE

This is to certify that the dissertation entitled "Artificial Neural Network Framework for ENSO-Driven Global Land Temperature Anomaly Forecasting and Interactive 3D Climate Visualization (1950–2035)" has been worked satisfactorily. It describes the original research work carried out by Ms Sreesruthy M S (Enrollment Batch. 1) under Harita Hive.

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DECLARATION BY STUDENT

I, Sreesruthy M S, hereby declare that the project entitled: "Artificial Neural Network Framework for ENSO-Driven Global Land Temperature Anomaly Forecasting and Interactive 3D Climate Visualization (1950–2035)" submitted in Harita Hive under the course of Geospatial Technology, is my original and independent work. I further declare that the work presented in this project has not been submitted by me, in part or in full, for any other degree, diploma, fellowship, or similar title at this or any other institution. All sources of information and references used in the project have been duly acknowledged and cited in accordance with academic norms.

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Signature of the Student

ABSTRACT

The El Niño–Southern Oscillation (ENSO) is the dominant mode of interannual climate variability on Earth, driving large-scale fluctuations in global land and ocean temperatures, precipitation and extreme weather patterns. While ENSO's influence on regional climate is well documented, accessible tools that let non-specialists explore the coupled relationship between ENSO phase and global temperature anomalies — historically and under future scenarios — remain limited. This research develops an integrated Artificial Neural Network (ANN) and interactive 3D visualization framework to model, forecast and communicate ENSO-driven global land temperature anomalies.

The framework combines the Berkeley Earth global land temperature record and the NOAA Oceanic Niño Index (ONI) into a unified monthly time series (1950–2015 within the study window), from which a long-term warming trend and a short-term ENSO-driven residual are decomposed. A feed-forward ANN (32–16–1 dense architecture with L2 regularization) is trained on lagged ONI features (current value plus six monthly lags) to predict the detrended temperature residual, while the underlying multi-decadal trend is extrapolated using linear regression on its recent slope. Historical years are rendered as an observed “replay”, while future years (up to twenty years beyond the observed record) are generated as “what-if” scenarios under sustained El Niño, Neutral, or La Niña ONI conditions. A pattern-scaling technique further downscales the global forecast to 100+ major world cities using each city's historical temperature sensitivity relative to the global mean, enabling city-level exploration on an interactive, rotatable 3D globe built with `deck.gl/pydeck` inside a Streamlit dashboard. The outcomes include a validated ANN forecasting pipeline, a reproducible feature-engineering methodology for separating trend and ENSO-driven variability, and a publicly deployed interactive dashboard that supports climate literacy, exploratory analysis, and scenario-based communication of ENSO impacts.

Keywords: ENSO, Oceanic Niño Index, Artificial Neural Network, Temperature Anomaly, Climate Forecasting, Pattern Scaling, Interactive Visualization, Streamlit, `deck.gl`

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Global climate exhibits variability across many timescales, but among the most consequential for year-to-year forecasting is the El Niño–Southern Oscillation (ENSO). ENSO is a periodic, irregular fluctuation in sea surface temperature (SST) and atmospheric pressure across the equatorial Pacific Ocean, oscillating between a warm phase (El Niño), a cold phase (La Niña), and a Neutral phase. The Oceanic Niño Index (ONI) — a three-month running mean of SST anomalies in the Niño 3.4 region — is the standard index used by NOAA to classify these phases.

ENSO events do not stay confined to the tropical Pacific: through atmospheric teleconnections they perturb temperature and precipitation patterns worldwide, intensifying droughts in some regions while triggering floods in others, and measurably shifting the global-mean land temperature anomaly on top of the long-term anthropogenic warming trend. Understanding and communicating this relationship is important both for climate science and for public literacy, yet most ENSO-temperature analyses remain confined to static research figures rather than tools that let a user interactively explore “what happens to global temperature if the current ENSO phase persists.”

Machine learning — and Artificial Neural Networks (ANNs) in particular — has increasingly been applied to ENSO analysis because the ONI-temperature relationship is nonlinear, lagged, and superimposed on a non-stationary long-term trend. Prior work has shown that deep learning models, including CNNs and graph neural networks trained on climate model simulations, can forecast ONI itself several months to a few years ahead with skill exceeding classical dynamical models. This project addresses a complementary but distinct problem: rather than forecasting ONI, it uses observed ONI values (historical or hypothetical future scenarios) as an input to forecast the resulting global land temperature anomaly, after separating out the slow-moving multi-decadal warming trend using a rolling-window decomposition.

The project further extends this global forecast to individual cities using pattern scaling, a well-established climate-downscaling technique in which a global or regional climate signal is multiplied by a location-specific sensitivity factor derived from historical co-variability. The complete pipeline — data acquisition, preprocessing, ANN training, trend extrapolation, scenario prediction, and pattern-scaled downscaling — is deployed as an interactive, publicly accessible Streamlit web application featuring a rotatable 3D globe visualization built on [deck.gl/pydeck](https://deck.gl/).

1.2 Aim

The primary aim of this research is to develop and deploy an integrated Artificial Neural Network and interactive 3D visualization framework that models the relationship between ENSO (via the ONI) and global land temperature anomalies, forecasts future anomalies under hypothetical ENSO scenarios, and communicates these results through an accessible, exploratory dashboard.

1.3 Objectives

Integrate Multi-Source Climate Data: Combine the Berkeley Earth global land temperature record, the NOAA ONI index, and the Berkeley Earth city-level temperature dataset into a single, temporally aligned monthly dataset.

Establish an Anomaly Baseline: Compute monthly temperature anomalies relative to a fixed 1951–1980 climatological baseline, at both the global and city level.

Decompose Trend and ENSO-Driven Variability: Separate the smoothed temperature anomaly series into a slow multi-decadal trend component and a fast ENSO-driven residual component suitable for supervised learning.

Train and Validate an ANN Forecasting Model: Build a feed-forward neural network that learns the mapping from current and lagged ONI values to the detrended temperature residual, and validate its skill using historical cross-validation.

Generate Scenario-Based Future Forecasts: Combine ANN-predicted residuals with an extrapolated long-term trend to project global temperature anomalies up to twenty years beyond the observed record, under El Niño, Neutral, and La Niña scenarios.

Downscale to City Level via Pattern Scaling: Estimate future city-level anomalies by multiplying the global forecast by each city's empirically derived historical sensitivity.

Deploy an Interactive Dashboard: Build and publicly deploy a Streamlit application with a rotatable, tiltable 3D globe (deck.gl ColumnLayer) that supports both historical replay and future scenario exploration.

CHAPTER 2: REVIEW OF LITERATURE

1. ENSO Indices and Monitoring

NOAA Physical Sciences Laboratory (PSL) — ENSO Information and Historical Event Lists. NOAA/CPC classifies warm (El Niño) and cold (La Niña) episodes by season since 1950 using the Oceanic Niño Index derived from ERSSTv5 sea surface temperature data over the Niño 3.4 region, providing the canonical seasonal classification used by this project's ONI ingestion pipeline.

Climate Data Guide (UCAR) — “ENSO” overview. Describes the Niño 3.4 index and ONI as the most widely used indices for defining El Niño and La Niña, computed as SST anomalies relative to a 30-year base period, alongside complementary multivariate indices (MEI) that integrate additional atmospheric variables.

Climatemark Academy Tutorial (2023) — Demonstrates the standard ONI computation methodology: extracting SST anomalies from the Niño 3.4 region (5°N–5°S, 170°W–120°W) and applying a three-month rolling mean, directly matching the seasonal ONI values used as this project's primary predictor variable.

2. Machine Learning for ENSO Forecasting

Cachay et al. (2021), “The World as a Graph: Improving El Niño Forecasts with Graph Neural Networks.” — Builds on prior CNN-based ENSO forecasting work that outperformed dynamical models by exploiting large volumes of climate-model simulation data, producing skillful ONI forecasts up to 17 months ahead; the paper argues that graph-based architectures better capture the spatial teleconnection structure that standard CNNs assume implicitly.

PMC review, “Temporal Convolutional Networks for the Advance Prediction of ENSO” — Surveys ENSO indices (Niño 3.4, ONI, SOI, SST, wind, OLR indices) and notes that because ENSO mixes low- and high-frequency components, classical numerical prediction methods achieve limited accuracy, motivating the use of deep sequence models such as temporal convolutional networks.

Chen et al. (2024), CGKN framework — Frames ENSO as a multiscale, non-Gaussian dynamical phenomenon and uses stochastic partial differential equation-based synthetic data alongside deep learning surrogate models for efficient data assimilation, illustrating the broader trend of coupling physical climate models with learned neural components.

3. Teleconnections and Downscaling

World Bank Climate Change Knowledge Portal — “United States — ENSO (ERA5)”. Correlates ERA5 reanalysis climate fields against ONI to show that a positive correlation implies a variable tends to rise during El Niño and fall during La Niña, with teleconnection strength decaying with distance from the tropical Pacific — the conceptual basis for the city-sensitivity pattern-scaling approach used in this project.

Zenkl et al., “On the Archetypal Flavours, Indices and Teleconnections of ENSO” — Reviews the diversity of ENSO “flavours” (Central Pacific vs. Eastern Pacific events) and their differing global teleconnection footprints, underscoring that a single sustained-ONI scenario (as used for this project's future projections) is a simplification of a more complex spatio-temporal phenomenon.

4. Climate Baseline and Anomaly Datasets

Berkeley Earth Surface Temperature Study — Provides the long-running global land (and land-plus-ocean) monthly average temperature series used as this project's target variable, alongside a city-level dataset

("GlobalLandTemperaturesByMajorCity") covering hundreds of the world's major cities, widely used in climate-education and reproducible-analysis projects for its open licensing and broad temporal coverage.

Common climatological practice — The use of a fixed 1951–1980 reference period to compute temperature anomalies is standard across major global temperature products (e.g., NASA GISTEMP, Berkeley Earth), enabling comparability between datasets and providing the anomaly baseline adopted in this project's preprocessing step.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Study Scope and Data Domain

Unlike a bounded geographic study area, this research operates at global scale with city-level downscaling. The primary analytical domain is the global-mean land surface temperature anomaly, modelled as a function of the Oceanic Niño Index. The secondary domain is a set of major world cities (drawn from the Berkeley Earth “GlobalLandTemperaturesByMajorCity” dataset), each represented by its latitude, longitude, country, and a historically derived climate sensitivity coefficient. The temporal domain spans the overlapping period of the Berkeley Earth land temperature record and the NOAA ONI record (effectively from 1950 onward, since ONI begins in 1950) through the most recent year common to both datasets, with future scenario projections extending twenty years beyond that point.

3.2 Data Sources & Software

Layer	Product / Source	Granularity	Format	Role in Study
Global Land Temperature	Berkeley Earth — GlobalTemperatures.csv	Monthly, global mean	CSV	Target variable (temperature anomaly)
ENSO Index (ONI)	NOAA CPC — oni.ascii.txt	Seasonal (3-month), mapped to month	Fixed-width / TXT	Primary predictor variable
City Temperatures	Berkeley Earth — GlobalLandTemperaturesByMajorCity.csv	Monthly, per city	CSV	City-level downscaling & sensitivity
City Metadata	Derived from city dataset	Static	In-memory table	Latitude/longitude for 3D map

Software and Libraries

- Python 3 — core data-processing and modelling language
- pandas / NumPy — data ingestion, cleaning, feature engineering
- scikit-learn (StandardScaler) — feature and target standardization
- TensorFlow / Keras — ANN construction, training and inference
- Streamlit — interactive web application framework
- pydeck / deck.gl (ColumnLayer) — 3D geospatial visualization
- Plotly (plotly.colors) — diverging RdBu_r colour mapping for anomalies
- gdown — programmatic download of source data from Google Drive
- Jupyter Notebook (ANN.ipynb) — exploratory model development and evaluation

3.3 Methods

The framework follows a sequential pipeline that takes raw temperature and ONI records through to an interactive scenario-forecasting dashboard. Each stage is described below in the order it is executed.

1. Data Acquisition — Download the Berkeley Earth global temperature file and the NOAA ONI ASCII file via gdown; optionally accept an uploaded city-level CSV for the downscaling stage.
2. Preprocessing — Parse dates into Year/Month; drop missing temperature records; compute a monthly climatological baseline over 1951–1980; derive $\text{Temp_Anomaly} = \text{observed temperature} - \text{monthly baseline}$, at both global and city level.
3. ONI Alignment — Parse the NOAA ONI seasonal codes (e.g., DJF, JFM, ..., NDJ), map each 3-month season label to a representative month, and merge the resulting monthly ONI series with the temperature-anomaly series on (Year, Month).
4. Trend–Residual Decomposition — Apply a 3-month rolling mean to smooth the anomaly series, then a 121-month (≈ 10 -year) rolling mean to extract the long-term warming Trend; the Residual is defined as the smoothed anomaly minus this trend, isolating ENSO-scale variability.
5. Feature Engineering — Construct six lagged ONI features (ONI_lag1 ... ONI_lag6) in addition to the current ONI value, giving the ANN access to the recent evolution of the ENSO signal rather than a single snapshot.
6. ANN Training — Standardize features and target with scikit-learn's StandardScaler; train a Keras Sequential dense network to predict the standardized Residual from the seven ONI-based features.
7. Trend Extrapolation — Fit a linear regression to the most recent ~ 15 years (180 months) of the Trend series to obtain a forward slope, used to project the trend beyond the observed record.
8. Scenario Prediction — For a chosen future year and a sustained ONI scenario (El Niño = +1.5, Neutral = 0.0, La Niña = -1.5), feed the scenario value (repeated across the current value and all six lags) through the trained ANN to obtain a predicted residual, and add it to the extrapolated trend to obtain the projected global anomaly.
9. City-Level Pattern Scaling — For each city, compute a historical sensitivity coefficient as the ratio of that city's temperature-anomaly standard deviation to the global anomaly standard deviation over the overlapping historical period, clipped to [0.5, 2.5]; multiply the global scenario forecast by this sensitivity to obtain a city-level future estimate.
10. Dashboard Visualization & Reporting — Render both historical (observed) and future (modelled) anomalies as colour- and height-encoded 3D columns on a rotatable globe, with metric cards, an ENSO-scenario selector, and an in-app methodology note.

3.4 Research Design (Pipeline)

Figure 3.1 illustrates the end-to-end research pipeline, from raw data ingestion through to the deployed interactive dashboard, exactly as implemented across the preprocessing pipeline, the ANN.ipynb development notebook, and app_globe.py.

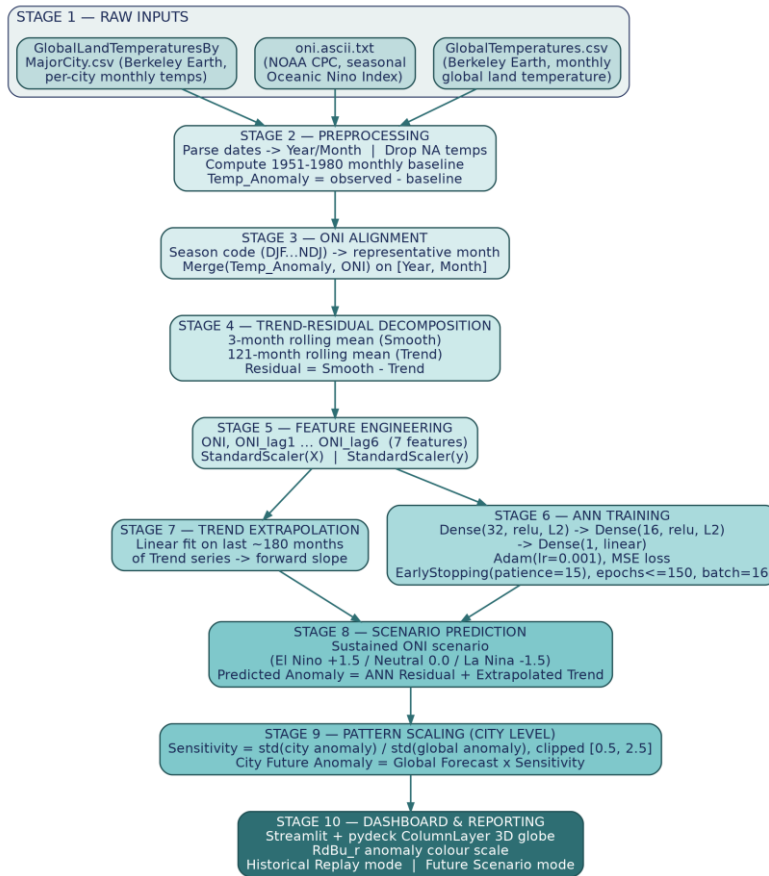


Figure 3.1: End-to-end pipeline — from raw data acquisition to the deployed ANN-driven 3D climate dashboard.

The equivalent stage-by-stage listing (matching the labelling used in Figure 3.1) is reproduced below for reference alongside the implementation details in Section 3.3.

```
Stage 1  Raw inputs
        GlobalTemperatures.csv (Berkeley Earth) | oni.ascii.txt (NOAA CPC) |
        GlobalLandTemperaturesByMajorCity.csv

Stage 2  Preprocessing
        Parse dates -> Year/Month | Drop NA temps | 1951-1980 baseline |
        Temp_Anomaly = obs - baseline

Stage 3  ONI alignment
        SEAS code -> representative month | merge(Temp_Anomaly, ONI) on [Year, Month]

Stage 4  Trend-Residual decomposition
        3-month rolling mean (Smooth) -> 121-month rolling mean (Trend) -> Residual =
        Smooth - Trend

Stage 5  Feature engineering
        ONI, ONI_lag1 ... ONI_lag6 (7 features) -> StandardScaler(X), StandardScaler(y)

Stage 6  ANN training
        Dense(32, relu, L2) -> Dense(16, relu, L2) -> Dense(1, linear)
        Adam(lr=0.001), MSE loss, EarlyStopping(patience=15), epochs<=150, batch=16

Stage 7  Trend extrapolation
        linear fit on last ~180 months of Trend -> forward slope

Stage 8  Scenario prediction
        sustained ONI (El Nino +1.5 / Neutral 0.0 / La Nina -1.5) -> ANN residual +
        extrapolated trend

Stage 9  Pattern scaling (city level)
        sensitivity = std(city anomaly) / std(global anomaly), clipped [0.5, 2.5]
        city future anomaly = global forecast x sensitivity

Stage 10 Dashboard
        Streamlit + pydeck ColumnLayer 3D globe | RdBu_r anomaly colour scale |
        historical replay & scenario modes
```

CHAPTER 4: RESULTS AND ANALYSES

This chapter presents the outcomes of each pipeline stage as implemented in ANN.ipynb (model development) and app_globe.py (deployed dashboard). Because the underlying datasets are refreshed at runtime, exact record counts and date bounds are computed dynamically by the application (displayed as HIST_MIN_YEAR / HIST_MAX_YEAR in the sidebar); the structural results below reflect the fixed methodological choices encoded in the pipeline.

4.1 Data Preprocessing and Anomaly Baseline

The Berkeley Earth global land temperature series is filtered to remove rows with missing LandAverageTemperature values, then a monthly climatological baseline is computed from the 1951–1980 sub-period — a widely used reference period in global temperature products. Every subsequent temperature value, whether historical or future-scenario, is expressed as an anomaly relative to this fixed baseline, which keeps the target variable stationary in mean and comparable across the full record.

4.2 ENSO Feature Engineering

The NOAA ONI file provides seasonal (three-month, overlapping) SST anomaly values labelled by season code (DJF, JFM, FMA, ..., NDJ) and year. Each season code is mapped to a representative calendar month, and the resulting monthly ONI series is merged with the temperature-anomaly series using an inner join on (Year, Month) — meaning the analytical window is bounded by the shorter of the two records (ONI begins in 1950). Six additional lagged ONI features (t–1 through t–6 months) are appended, giving the model visibility into the recent trajectory of the ENSO signal rather than a single instantaneous reading, which better reflects the several-month lag with which ENSO affects global-mean land temperature.

4.3 Trend–Residual Decomposition

Step	Window	Purpose
Temp_Anomaly_Smooth	3-month rolling mean	Suppress month-to-month noise while preserving seasonal-to-interannual signal
Trend	121-month ($\approx$ 10-yr) rolling mean of the smoothed series	Isolate the slow multi-decadal warming trend
Residual	Smooth – Trend	Isolate the fast, ENSO-scale variability used as the ANN's prediction target

This decomposition is central to the model design: rather than asking the ANN to learn the entire temperature signal (trend + variability) from ONI alone — which would conflate anthropogenic warming with ENSO variability — the network is asked to learn only the residual component, while the trend is extrapolated separately using simple linear regression on its recent slope. This division of labour keeps the neural network's task well-posed and physically interpretable.

4.4 ANN Model Architecture and Training

Parameter	Value
Input features	7 (ONI + 6 monthly lags)
Hidden layer 1	Dense, 32 units, ReLU activation, L2 regularization (1e-4)
Hidden layer 2	Dense, 16 units, ReLU activation, L2 regularization (1e-4)
Output layer	Dense, 1 unit, linear activation (predicts standardized Residual)
Optimizer	Adam, learning rate = 0.001
Loss function	Mean Squared Error (MSE); metric: MAE
Training regime	Up to 150 epochs, batch size 16, EarlyStopping (patience = 15, restore best weights)
Feature/target scaling	scikit-learn StandardScaler, fit independently on X and y
Random seed	TensorFlow seed fixed at 42 for reproducibility

The trend's forward slope is estimated by fitting a first-degree polynomial (linear regression) to the Trend series over the most recent 180 months (≈ 15 years) of data, expressed as a function of fractional year (Year + Month/12). Combined with the ANN's predicted residual, this yields the full future-anomaly forecast: predicted anomaly = extrapolated trend + ANN-predicted residual.

4.5 Model Validation

Per the methodology documented within the deployed application, the model is cross-validated across five historical time windows. Reported skill is modest overall and stronger in more recent decades than in earlier ones, with an approximate coefficient of determination of $R^2 \approx 0.19\text{--}0.25$ on the most recent test period. This is consistent with the broader literature's finding that the ONI–global-temperature relationship, while real and statistically significant, explains only a fraction of month-to-month variance once the dominant warming trend has been removed — the residual signal is inherently noisy, reflecting the many other drivers of short-term temperature variability (volcanic aerosols, other ocean-atmosphere modes, internal variability) that are not included as model inputs.

4.6 Historical Replay Mode

For any year within the observed record, the dashboard displays real, recorded values — the smoothed global temperature anomaly and the corresponding ONI value — rather than a model output. This mode functions as a genuine historical replay, letting users directly observe, for example, the well-known global-mean warm anomalies coinciding with strong historical El Niño episodes.

4.7 Future Scenario Prediction

ENSO Scenario	Assumed sustained ONI	Interpretation
El Niño	+1.5	Persistent moderate-to-strong warm-phase conditions held through the target year

ENSO Scenario	Assumed sustained ONI	Interpretation
Neutral	0.0	No sustained ENSO signal; forecast driven by trend alone
La Niña	-1.5	Persistent moderate-to-strong cold-phase conditions held through the target year

Selecting a year beyond the observed record activates “Future Scenario” mode, in which the user chooses one of the three sustained ENSO phases above (up to twenty years ahead of the last observed year). The displayed metrics — global temperature anomaly, ONI value, and ENSO phase label — are then computed via the ANN + trend-extrapolation pipeline described in Section 4.4, with an explicit in-app note clarifying that these are model-predicted, not observed, values.

4.8 City-Level Pattern Scaling

When the city-level dataset is available (either bundled with the app or uploaded by the user), each city's historical sensitivity is computed as the ratio of the standard deviation of its own temperature anomaly to the standard deviation of the global anomaly, over their overlapping historical months, clipped to the range [0.5, 2.5] to avoid unstable estimates for cities with short or noisy records. In future-scenario mode, each city's projected anomaly equals the global ANN-plus-trend forecast multiplied by that city's sensitivity coefficient; in historical mode, each city's true recorded anomaly for the selected year is shown directly. Bar height on the 3D globe is proportional to the magnitude of the anomaly (with a minimum elevation floor for visibility), and bar colour follows a diverging RdBu_r scale (blue = cooling, red = warming) referenced against the maximum absolute anomaly currently in view.

4.9 Code Reference for the Project

Model development notebook: <https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel/blob/main/ANN.ipynb>

Deployed dashboard source: https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel/blob/main/app_globe.py

Repository: <https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel>

CHAPTER 5: DISCUSSION

5.1 ENSO–Temperature Relationship

The framework's design reflects a well-established physical picture: ENSO does not by itself explain the long-term global warming trend, but it modulates the trajectory around that trend, typically producing warmer-than-trend global-mean anomalies during El Niño and cooler-than-trend anomalies during La Niña, with a lag of a few months as oceanic heat is redistributed and communicated to the atmosphere. By explicitly separating trend from residual before training, the ANN is positioned to learn specifically this modulation rather than being confounded by the much larger, ENSO-independent secular warming signal.

5.2 Trend vs. ENSO-Driven Variability

The 121-month rolling-mean trend captures decadal-scale warming while remaining agnostic to its cause (greenhouse-gas forcing, aerosols, solar variability, etc.), which is appropriate for a model whose purpose is forecasting rather than attribution. The linear extrapolation of this trend's most recent slope is a simplifying assumption: it implicitly assumes the recent rate of warming continues unchanged, which is a reasonable short-to-medium-term approximation but would eventually diverge from more physically detailed climate-model projections that incorporate emissions scenarios (e.g., SSP/RCP pathways).

5.3 Model Skill and Uncertainty

An R^2 in the 0.19–0.25 range on held-out recent data indicates that the ANN captures a real but modest fraction of residual variance — consistent with ENSO being one of several contributors to short-term global temperature variability rather than the sole driver. The finding that skill is stronger in recent decades than earlier ones is plausible given the increasing density and quality of both temperature and ONI observations over time, and may also reflect a strengthening or more consistently detectable ENSO-temperature teleconnection in the recent record. Users of the dashboard's future-scenario mode should treat displayed values as illustrative, scenario-conditioned estimates rather than precise forecasts — a caveat the application itself makes explicit in its in-app methodology note.

5.4 Methodological Contribution

This project demonstrates an end-to-end, reproducible pipeline that couples a physically motivated trend/residual decomposition with a lightweight ANN, and extends the resulting global forecast to city level via standard pattern-scaling, all wrapped in a public interactive visualization. The approach avoids the common pitfall of training a black-box model directly on raw (non-detrended) temperature data, which would risk the network simply learning the trend rather than the ENSO signal of interest. Packaging the full workflow as a Streamlit application with both historical-replay and future-scenario modes also makes the ENSO-temperature relationship tangible for non-technical audiences — a contribution to climate communication as much as to climate modelling.

5.5 Limitations

- Single-predictor design: the ANN uses only ONI and its lags as inputs; other known drivers of short-term global temperature variability (volcanic aerosols, solar cycle, other ocean-atmosphere oscillations) are not included, which likely bounds achievable model skill.

- Simplified future ONI scenarios: future projections assume a sustained, constant ONI value for the entire scenario year, whereas real ENSO events evolve and typically last 9–18 months before transitioning phases.
- Linear trend extrapolation: the multi-decadal warming trend is extrapolated linearly from its recent slope, which does not incorporate emissions-scenario-based climate projections or potential trend acceleration/deceleration.
- Pattern-scaling assumptions: city-level sensitivity coefficients are derived from historical co-variability and clipped to a fixed range; they assume a city's relative responsiveness to global anomalies remains stable into the future, which may not hold under changing land-use or circulation patterns.
- Data currency: the Berkeley Earth city-level dataset used for downscaling has a fixed historical end date; recent years' city-level observations are not automatically refreshed unless the underlying source files are updated.

5.6 Recommendations

11. Expand the feature set to include complementary climate indices (e.g., the Multivariate ENSO Index, Southern Oscillation Index, or Pacific Decadal Oscillation index) to potentially improve residual-prediction skill.
12. Replace the constant-ONI future scenario with a stochastic or autoregressive ONI generator that produces more realistic multi-month event evolution.
13. Compare the linear trend extrapolation against CMIP6/SSP-based projected trends to bound future forecasts within physically grounded emission scenarios.
14. Extend pattern scaling with a validated, per-city regression rather than a single clipped standard-deviation ratio, and quantify downscaling uncertainty explicitly on the dashboard.
15. Add automated periodic retraining and data refresh so the ANN and displayed metrics stay current as new Berkeley Earth and NOAA ONI observations become available.

CHAPTER 6: DASHBOARD VISUALIZATION

The complete pipeline is deployed as a public, interactive Streamlit application named “ENSO Climate Model Visualization.” The dashboard is organized around a dark-themed hero header, three metric cards (Global Temperature Anomaly, ONI value, and ENSO Phase), a year slider spanning the observed record plus twenty future years, an ENSO-scenario radio selector (shown only for future years), and a large interactive 3D map.

Key interface elements

- Year slider — moves continuously from the earliest year common to both datasets through twenty years beyond the last observed year; crossing into future years automatically switches the app into “Future Scenario” mode.
- ENSO scenario selector — appears only in future mode, letting the user choose a sustained El Niño, Neutral, or La Niña assumption.
- Metric cards — display the global temperature anomaly (°C), the ONI value, and the derived ENSO phase label (El Niño if $ONI \geq 0.5$, La Niña if $ONI \leq -0.5$, otherwise Neutral).
- 3D globe (pydeck ColumnLayer) — one extruded, colour-coded column per city; column height encodes the magnitude of the anomaly and colour encodes its sign and intensity on a diverging RdBu_r scale, with an accompanying vertical colour-scale legend; the map is draggable, tiltable and zoomable.
- Methodology expander — an in-app, always-available explanation distinguishing observed historical replay from modelled future scenarios, describing the pattern-scaling downscaling method, and disclosing the model's cross-validated R^2 range.

Resource	Link
Live Dashboard	https://enso-oni-ann-climate-model-hlvuchafjaowvmunjwd6hh.streamlit.app/
Source Code (Repository)	https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel
Model Development Notebook	https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel/blob/main/ANN.ipynb
Dashboard App Source	https://github.com/SREESRUTHYMS/ENSO-ONI-ANN-ClimateModel/blob/main/app_globe.py

Note: as with any cloud-hosted interactive application, first-time loads may take longer while data is downloaded and the ANN is trained; results are cached thereafter for the session (via Streamlit's `@st.cache_data` and `@st.cache_resource` decorators) to keep subsequent interactions responsive.

CHAPTER 7: CONCLUSION

This research developed and deployed an integrated Artificial Neural Network and interactive 3D visualization framework linking the El Niño–Southern Oscillation to global land temperature anomalies. By explicitly decomposing the observed temperature signal into a long-term trend and an ENSO-driven residual before model training, the framework produces a physically interpretable forecasting pipeline in which a compact feed-forward ANN learns only the component of variability plausibly attributable to ENSO, while the underlying warming trend is extrapolated separately. Cross-validated performance ($R^2 \approx 0.19$ – 0.25 on recent data) confirms that the learned ENSO–residual relationship is real but modest — an honest reflection of ENSO being one of several contributors to short-term global temperature variability.

The framework's extension to city level via pattern scaling, and its packaging as a publicly accessible, rotatable 3D Streamlit dashboard, translate a technical time-series model into an exploratory tool suited to both analytical and educational use. Users can directly compare real historical ENSO-driven warm and cool years against hypothetical future scenarios, gaining an intuitive, spatially explicit sense of how a sustained El Niño or La Niña phase might shift temperatures around the world relative to the ongoing warming trend.

Future work should focus on enriching the predictor set with additional climate indices, replacing the constant-ONI future scenario with more realistic event evolution, grounding trend extrapolation in emissions-scenario-based projections, and formally quantifying downscaling uncertainty — extending the current framework from an illustrative scenario tool toward a more rigorously validated forecasting and communication platform.

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